RELATIVE ASSOCIATED PRIMES OVER AN AFFINE CHART DETECT RELATIVE ASSOCIATED POINTS OF THE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-relative-assassin-affine-open is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 1078-1095 and proof span 1097-1117 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the relative associated primes by the author's notation Ass_{A/R}(M) and to the sheaf version by Ass_{X/S}(F); both are preserved.

Editorial note. The statement's final sentence is an equivalence whose converse needs finite generation of the fibre prime; the proof marks that openly, and the hypothesis 'all fibres locally Noetherian' in the statement is what supplies it.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The definition of the relative associated primes of a module over a ring map, cited from the algebra chapter. Cited at proof lines 1097-1098. Names the algebraic set that the statement compares with the relative associated points of the sheaf.
- The comparison between relative associated primes of a module and associated primes of its base change to the residue field, cited from the algebra chapter. Cited at proof lines 1105-1107. Turns the algebraic condition into a condition on the fibre of the module, which is where the geometric statement lives.
- The same-chapter description of associated points of a quasi-coherent sheaf on an affine open by the associated primes of the module of sections. Cited at proof lines 1111-1112, this being the lane's card L4023 of this source. Converts the fibre condition into the statement that the point is associated to the restriction of the sheaf to the fibre.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 05AT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6861–6861. Same-chapter reference used in the retained author text at source line(s) 1052, 1099. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 05GB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7055–7055. Same-chapter reference used in the retained author text at source line(s) 1109. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02OK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3519–3519. Same-chapter reference used in the retained author text at source line(s) 168, 186, 423, 852, 1114. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $U \subset X$ and $V \subset S$ be affine opens with $f(U) \subset V$. Write $U = \mathrm{Spec}(A)$, $V = \mathrm{Spec}(R)$, and set $M = \Gamma(U, \mathcal{F})$. Let $x \in U$, and let $\mathfrak{p} \subset A$ be the corresponding prime. Then*

$$\mathfrak{p} \in Ass_{A/R}(M) \Rightarrow x \in Ass_{X/S}(\mathcal{F})$$

If all fibres X_s of f are locally Noetherian, then $\mathfrak{p} \in Ass_{A/R}(M) \Leftrightarrow x \in Ass_{X/S}(\mathcal{F})$ for all pairs $(\mathfrak{p}, x)$ as above.

Proof. The set $\mathrm{Ass}_{A/R}(M)$ is defined in Algebra, Definition 05AT. Choose a pair $(\mathfrak{p}, x)$. Let $s = f(x)$. Let $\mathfrak{r} \subset R$ be the prime lying under $\mathfrak{p}$, i.e., the prime corresponding to s. Let $\mathfrak{p}' \subset A \otimes_R \kappa(\mathfrak{r})$ be the prime whose inverse image is $\mathfrak{p}$, i.e., the prime corresponding to x viewed as a point of its fibre X_s. Then $\mathfrak{p} \in \mathrm{Ass}_{A/R}(M)$ if and only if $\mathfrak{p}'$ is an associated prime of $M \otimes_R \kappa(\mathfrak{r})$, see Algebra, Lemma 05GB. Note that the ring $A \otimes_R \kappa(\mathfrak{r})$ corresponds to U_s and the module $M \otimes_R \kappa(\mathfrak{r})$ corresponds to the quasi-coherent sheaf $\mathcal{F}_s|_{U_s}$. Hence x is an associated point of $\mathcal{F}_s$ by Lemma 02OK. The reverse implication holds if $\mathfrak{p}'$ is finitely generated which is how the last sentence is seen to be true. □